

Stock Market Fundamentals Index

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Abstract

We introduce the Market Fundamentals Index (MFI), a novel metric constructed from a tensor representation of firm-level accounting data and smoothed via Tucker decomposition. The MFI captures time-varying dispersion in market fundamentals, and we demonstrate its strong statistical dependence on a price-based financial chaos index (FCIX). Building on this tensor framework, we develop a predictive pipeline that imputes missing fundamentals through mask-aware Tucker decomposition, assembles quarterly lookback tensors, and forecasts next-quarter values via low-rank CANDECOMP/PARAFAC regression with firm-by-feature fixed effects and RMS normalization. We evaluate the approach against a per-feature ridge regression baseline using time-series cross-validation and a held-out test set.

Keywords: multilinear tensor regression, Tucker decomposition, CANDECOMP/PARAFAC, market fundamentals, volatility index, fixed effects, time-series forecasting

1 Introduction

This paper investigates asset price fluctuations and their relationship with underlying firm fundamentals. Incorporating fundamental accounting data is essential for capturing the full scope of publicly available market information. Together with news-based uncertainty measures and historical prices, these three information sources form the basis for rigorous tests of the efficient market hypothesis across multiple time horizons.

The practical significance is direct: if the market exhibits inefficiency at daily, monthly, or quarterly frequencies, investors can leverage historical data to forecast future asset trajectories. Conversely, if the market proves efficient, prediction-based

models must give way to alternative asset allocation strategies. Our framework therefore provides actionable guidance for traders and portfolio managers seeking to position investments optimally.

1.1 Notations and Preliminaries

Throughout this paper, scalar quantities, vectors, and matrices are denoted by lowercase lightface (e.g., x), lowercase boldface (e.g., $\mathbf{x}$), and uppercase boldface (e.g., $\mathbf{X}$) letters, respectively. All vectors are assumed to be column vectors unless stated otherwise. Boldface calligraphic letters (e.g., $\mathcal{X}$) are reserved for tensors, while their lightface calligraphic counterparts (e.g., $\mathcal{X}$) typically denote induced sample spaces.

Tensors are algebraic structures that generalize matrices to higher dimensions. The number of dimensions of a tensor is referred to as its order or mode. For instance, $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ denotes an N th order tensor with $K = \prod_{n=1}^N I_n$ elements in total. Analogous to matrix rows and columns, the mode- n fibers of a tensor are obtained by fixing all indices except the n th one. Fixing all but two indices yields the slices of the tensor.

The Frobenius norm of a tensor $\mathcal{X}$ is defined by

$$\|\mathcal{X}\|_F = \sqrt{\langle \mathcal{X}, \mathcal{X} \rangle} = \sqrt{\sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \dots \sum_{i_N=1}^{I_N} x_{i_1 i_2 \dots i_N}^2}. \quad (1)$$

To reduce tensor order and facilitate matrix computations, one often resorts to unfolding (also called matricization). The vector unfolding of $\mathcal{X}$, denoted by

$$\mathbf{x} = \text{vec}(\mathcal{X}) \in \mathbb{R}^K, \quad (2)$$

reshapes the tensor into a vector. The mode- n matricization of $\mathcal{X}$, given by

$$\mathcal{X}_{(n)} = [\mathbf{f}_1^{(n)}, \mathbf{f}_2^{(n)}, \dots, \mathbf{f}_{K/I_n}^{(n)}], \quad (3)$$

stacks the mode- n fibers as columns of a matrix of size $I_n \times (K/I_n)$.

The mode- n product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n}$ is denoted by $\mathcal{X} \times_n \mathbf{U}$ and defined element-wise as

$$(\mathcal{X} \times_n \mathbf{U})_{i_1 \dots i_{n-1} j i_{n+1} \dots i_N} = \sum_{i_n=1}^{I_n} x_{i_1 \dots i_N} u_{j i_n}, \quad (4)$$

yielding a tensor of dimensions $I_1 \times \dots \times I_{n-1} \times J \times I_{n+1} \times \dots \times I_N$. This operation is equivalent to multiplying $\mathbf{U}$ from the left on each mode- n fiber as follows:

$$\mathcal{Y} = \mathcal{X} \times_n \mathbf{U} \iff \mathcal{Y}_{(n)} = \mathbf{U} \mathcal{X}_{(n)}. \quad (5)$$

When dealing with third-order tensors $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, a useful perspective is to treat them as stacks of frontal slices $\mathcal{X}^{(n)} \in \mathbb{R}^{I_1 \times I_2}$ for $n = 1, 2, \dots, I_3$. Define the

operator

$$\text{unfold}(\mathcal{X}) := [\mathcal{X}^{(1)}, \mathcal{X}^{(2)}, \dots, \mathcal{X}^{(I_3)}]^\top, \quad (6)$$

which stacks all frontal slices vertically. The corresponding inverse operator **fold** satisfies

$$\text{fold}(\text{unfold}(\mathcal{X})) := \mathcal{X}. \quad (7)$$

Two widely used tensor decompositions are the Tucker and the polyadic (or CP) decompositions. The Tucker decomposition approximates a third-order tensor $\mathcal{X}$ by a dense core tensor $\mathcal{H} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ and factor matrices $\mathbf{A} \in \mathbb{R}^{I_1 \times R_1}$, $\mathbf{B} \in \mathbb{R}^{I_2 \times R_2}$, and $\mathbf{C} \in \mathbb{R}^{I_3 \times R_3}$ as follows:

$$\mathcal{X} \approx \hat{\mathcal{X}} = \mathcal{H} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{C}. \quad (8)$$

Alternatively, the polyadic decomposition expresses the tensor as a sum of rank-one components,

$$\mathcal{X} \approx \hat{\mathcal{X}} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r, \quad (9)$$

where the symbol $\circ$ denotes the vector outer product and R is the rank of the approximation. When R is the minimal number of components required to achieve equality, the decomposition is referred to as the canonical polyadic decomposition (CPD).

Finally, a third-order tensor $\mathcal{X}$ is said to be rank-one if it admits the representation

$$\mathcal{X} = \mathbf{a} \circ \mathbf{b} \circ \mathbf{c}, \quad (10)$$

for some vectors $\mathbf{a} \in \mathbb{R}^{I_1}$, $\mathbf{b} \in \mathbb{R}^{I_2}$, and $\mathbf{c} \in \mathbb{R}^{I_3}$.

2 Encoding Fundamentals Information

In this section, we develop a methodology for embedding stock market fundamentals into structured representations and identifying a subset of features that preserves the latent economic structure of the market. The ultimate objective is to facilitate the construction of a volatility index grounded in fundamental information. The selected features are required to align with a reference taxonomy of economic activity, namely the Global Industry Classification Standard (GICS), thus providing a bridge between qualitative categorization and quantitative representation.

2.1 Feature Selection via Clustering

Let $S = \{S_1, S_2, \dots, S_N\}$ denote the collection of assets observed in the market, and let $F = \{f_1, f_2, \dots\}$ denote the universe of available fundamental features. For each asset S_i , let $\mathbf{d}_i \in \mathbb{R}^{|F'|}$ denote the vector of observed values for a subset $F' \subset F$.

These vectors are assembled into a design matrix $\mathbf{D} \in \mathbb{R}^{N \times |F'|}$ defined as

$$\mathbf{D} = \begin{bmatrix} \mathbf{d}_1^\top \\ \mathbf{d}_2^\top \\ \vdots \\ \mathbf{d}_N^\top \end{bmatrix} = \begin{matrix} & \begin{matrix} f_1^* & f_2^* & \dots & f_{|F'|}^* \end{matrix} \\ \begin{matrix} s_1 \\ s_2 \\ \vdots \\ s_N \end{matrix} & \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1|F'|} \\ d_{21} & d_{22} & \dots & d_{2|F'|} \\ \vdots & \vdots & \ddots & \vdots \\ d_{N1} & d_{N2} & \dots & d_{N|F'|} \end{bmatrix} \end{matrix}.$$

Our objective is to identify a subset $F^* \subset F$ whose induced clustering structure naturally mirrors the established GICS sub-industry classification. Formally, let $\mathcal{C}(F')$ denote the number of clusters inferred from $\mathbf{D}$ via a clustering algorithm, and let $\mathcal{G}$ denote the cardinality of the GICS sub-industry partition (i.e., $\mathcal{G} = 158$). The optimal subset is then

$$F^* = \arg \min_{F' \subset F} |\mathcal{C}(F') - \mathcal{G}|. \quad (11)$$

This defines our best-match criterion, which balances parsimony in feature selection with fidelity to external economic structure. A common alternative to this discrete-matching approach is to measure partition similarity using adjusted Rand index or normalized mutual information, though such refinements are omitted here in favor of interpretability.

Traditional clustering algorithms such as k -means require the number of clusters to be specified a priori, which is incompatible with our goal of estimating structure from the data itself. To address this, we employ the affinity propagation algorithm [1], which treats every observation as a potential cluster exemplar and seeks to minimize an energy function through iterative message passing. This framework automatically determines the number of clusters, with control exerted through a tunable preference parameter. Technical details and convergence analysis are provided in [2].

An additional challenge lies in the representation of temporal information. The matrix $\mathbf{D}$ discards the temporal dynamics of each fundamental feature by aggregating values into a single statistic per asset. While this introduces a form of statistical coarsening, it is justified by the relative stationarity of taxonomic classification over short horizons. Sector and sub-industry affiliations typically undergo redefinition on a 4-5 year timescale, often in response to shifts in macroeconomic conditions or technological innovation. To reflect this, we construct $\mathbf{D}$ by averaging quarterly values of each feature from the first quarter of 2016 to the fourth quarter of 2019.¹

For each candidate subset $F' \subset F$, the design matrix $\mathbf{D}$ is constructed and subjected to affinity propagation clustering. The resulting number of clusters is then compared to $\mathcal{G} = 158$. The subset F^* is selected according to the best-match criterion. While dimensionality reduction methods such as PCA, Lasso, or autoencoders could be considered, we intentionally avoid these to preserve economic interpretability of the features. Moreover, robustness checks revealed that F^* remained stable across perturbed data subsets, indicating selection consistency.

In total, over 600 distinct quarterly and annual features were considered. Through extensive search and evaluation, a 40-feature subset F^* was identified, as detailed

¹Fundamental data retrieved from Compustat via the WRDS platform.

Table 1: Selected fundamental features used for constructing the design matrix.

Feature	Feature
Acquisitions	Inventories - Total
Assets - Other - Total	Investing Activities - Net Cash Flow
Assets and Liabilities	Investing Activities - Other
Capital Expenditures	Long-Term Debt - Total
Cash and Short-Term Investments	Long-Term Debt - Issuance
Cash Dividends	Liabilities Netting Other Adjustments
Common/Ordinary Equity - Total	Noncontrolling Interest
Comprehensive Income - Total	Non-Operating Income (Expense) - Total
Cost of Goods Sold	Operating Activities - Net Cash Flow
Debt in Current Liabilities	Operating Income Before Depreciation
Earnings Per Share (Basic)	Preferred/Preference Stock (Capital) - Total
Earnings Per Share (Diluted)	Pretax Income
Excess Tax Benefit of Stock Options	Receivables - Total
Extraordinary Items	Sale of Common and Preferred Stock
Financing Activities - Net Cash Flow	Sale of Investments
Funds from Operations - Other	Sale of PPE and Investments - Gain/Loss
Quarterly Income Before Extraordinary Items	Sales/Turnover (Net)
Annual Income Before Extraordinary Items	Short-Term Investments - Total
Income Taxes	Special Items
Intangible Assets - Total	Stockholders Equity

in Table 1. The affinity propagation algorithm applied to $\mathbf{D}$ constructed from F^* yielded 156 clusters, just two fewer than the GICS-defined sub-industry count. This near match supports the claim that F^* provides a structurally faithful representation of economic heterogeneity across the market.

2.2 Market Fundamentals Index

We now introduce a tensor-based volatility index that summarizes the aggregate dynamics of stock market fundamentals over time. This Market Fundamentals Index (MFI) parallels the FCIX developed in prior literature, but it is uniquely derived from fundamental accounting data rather than asset returns.

Let $\mathcal{D} \in \mathbb{R}^{N \times |F^*| \times |\mathcal{T}|}$ denote a third-order design tensor storing the temporal trajectories of selected fundamental features for N assets, where $|\mathcal{T}|$ is the number of time periods. In the empirical construction below, $\mathcal{D}$ contains $N = 496$ firms, $|F^*| = 40$ quarterly accounting features, and $|\mathcal{T}| = 140$ quarters from 1990Q1 through 2024Q4. The structure of this tensor is illustrated in Figure 1.

To characterize the structural complexity of $\mathcal{D}$ and inform the construction of MFI, we begin by applying a polyadic decomposition as follows:

$$\mathcal{D} \approx \hat{\mathcal{D}} = \sum_{r=1}^R \mathbf{s}_r \circ \mathbf{f}_r \circ \mathbf{t}_r, \quad (12)$$

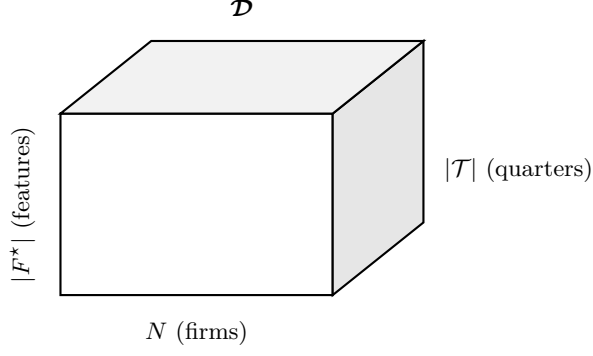


Fig. 1: Schematic of a third-order market fundamentals design tensor.

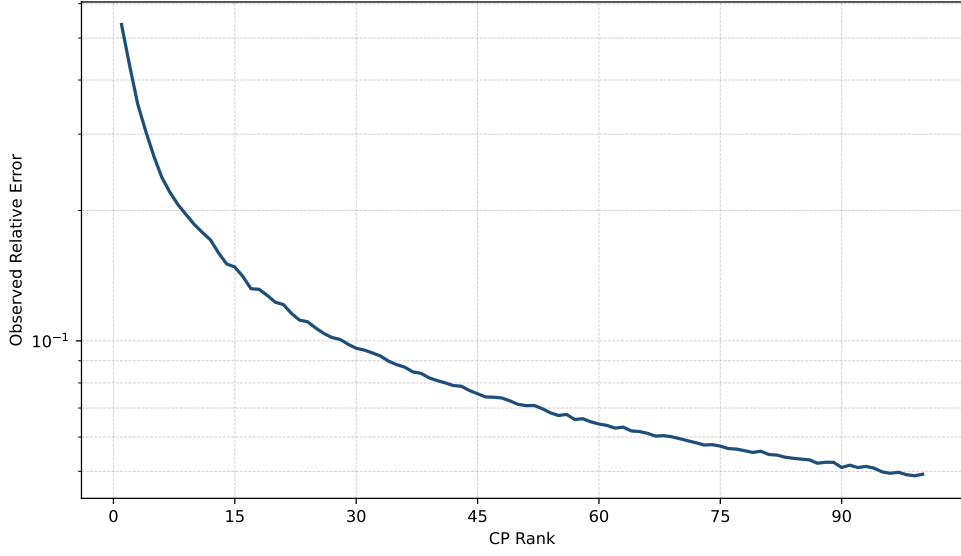


Fig. 2: Plot of the relative error of polyadic decomposition of $\mathcal{D}$ for different values of R .

where $\mathbf{s}_r \in \mathbb{R}^N$, $\mathbf{f}_r \in \mathbb{R}^{|F^*|}$, and $\mathbf{t}_r \in \mathbb{R}^{|\mathcal{T}|}$ denote the r th rank-one component. As shown in Figure 2, the relative error of approximation remains significantly high even for $R = 100$, indicating that $\mathcal{D}$ is not well approximated by low-rank additive components. This suggests that the interactions among stocks, fundamentals, and time exhibit non-separable and highly nonlinear structure.

Motivated by this observation, we instead employ Tucker decomposition, which generalizes matrix SVD to higher-order tensors by introducing a core tensor to capture

multiway interactions. Specifically, we write

$$\mathcal{D} \approx \widehat{\mathcal{D}} = \mathcal{H} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T}, \quad (13)$$

where $\mathcal{H} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ is the core tensor, and $\mathbf{S} \in \mathbb{R}^{N \times R_1}$, $\mathbf{F} \in \mathbb{R}^{|F^*| \times R_2}$, and $\mathbf{T} \in \mathbb{R}^{|\mathcal{T}| \times R_3}$ are the factor matrices. The multilinear rank vector is defined as $\mathbf{R} = [R_1, R_2, R_3]^\top$. The mask-aware Tucker decomposition is obtained by solving the observed-entry optimization problem

$$\min_{\mathcal{H}, \mathbf{S}, \mathbf{F}, \mathbf{T}} \|\mathbf{M} \odot (\mathcal{D} - \mathcal{H} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T})\|_F, \quad (14)$$

where $\mathbf{M}$ is the binary observation mask and $\odot$ denotes elementwise multiplication.

To preserve economic interpretability, we set R_1 equal to the number of observed GICS industry codes in the expanded sample and retain the full feature dimension in the second mode. This yields $R_1 = 67$ and $R_2 = 40$. We then fix $R_3 = 20$ to summarize the dominant temporal modes and evaluate the observed-entry relative approximation error defined by

$$\text{rel}_{\text{error}} = \frac{\|\mathbf{M} \odot (\mathcal{D} - \widehat{\mathcal{D}})\|_F}{\|\mathbf{M} \odot \mathcal{D}\|_F}. \quad (15)$$

Rather than minimizing this error to the lowest achievable level, we intentionally select a moderate value of R_3 that filters out noise and captures only the most persistent components. Based on our empirical evaluation, we use the multilinear rank vector $[67, 40, 20]$, which yields an observed-entry relative approximation error of 22.2%.

The temporal factor matrix $\mathbf{T}$ now contains R_3 latent time series of length $|\mathcal{T}|$. Each column captures a distinct latent mode of variation in the temporal dimension of the tensor. To aggregate these into a single index, we define the Market Fundamentals Index by taking the average magnitude across the R_3 components at each time t , whose formal definition is provided below.

Definition 1 (MFI). Let $\mathcal{D}$ be a third-order tensor of market fundamentals smoothed via Tucker decomposition as $\widehat{\mathcal{D}} = \mathcal{H} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T}$, with $\mathbf{T} \in \mathbb{R}^{|\mathcal{T}| \times R_3}$. The Market Fundamentals Index (MFI) is defined by

$$\text{MFI}(t) := \frac{1}{R_3} \sum_{k=1}^{R_3} |\mathbf{T}_{tk}|, \quad (16)$$

for $t = 1, \dots, |\mathcal{T}|$.

The absolute value ensures that phase shifts or sign reversals do not cancel signal strength, and the averaging operation yields a smoothed, aggregate representation of the total volatility embedded in the fundamentals. As in the case of FCIX_t , we denote the time series realization of this statistic by MFI_t .

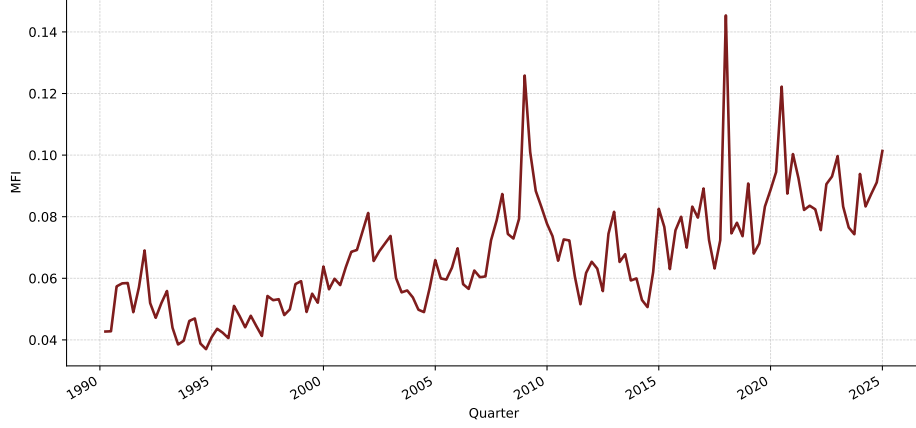


Fig. 3: Plot of the quarterly MFI over the period January 1990–December 2024.

Figure 3 displays the quarterly realizations of MFI from January 1990 through December 2024. The time series reveals distinct structural episodes, marked by substantial shifts in the volatility of firm-level fundamentals. Notably, the index remains relatively stable from 1990 through the early 2000s, followed by a sharp regime break in 2002-2003, after which a sustained increase in MFI levels is observed. A pronounced spike appears during the 2007-2009 financial crisis, peaking near the apex of the 2008 global meltdown. The elevated levels in this period reflect heightened dispersion and instability in accounting fundamentals, consistent with widespread balance sheet stress across sectors.

Post-crisis, the MFI declines steadily, stabilizing around 2013, where it enters a low-volatility regime with relatively minor fluctuations. A modest upward drift is again visible beginning in 2018, potentially foreshadowing structural shifts leading into the 2020 pandemic period. These patterns suggest that MFI effectively captures systemic dislocations in fundamental data and exhibits temporal alignment with major financial crises. The dynamics of this index and its potential predictive content are further explored in subsequent empirical tests.

3 Statistical Dependence Between MFI and FCIX

We now examine the empirical relationship between the MFI and the FCIX, with the goal of determining whether their co-movement reflects genuine statistical dependence. Identifying such dependence would support the hypothesis that fluctuations in market fundamentals are structurally linked to volatility in asset prices, thereby providing empirical grounding for the efficient market hypothesis explored in the following sections.

We begin by computing the sample cross-correlation between the quarterly time series $\{MFI_t\}$ and $\{FCIX_t\}$ over a range of lags. Figure 4 illustrates the resulting correlation coefficients from -10 to $+10$ quarters. The strongest association is

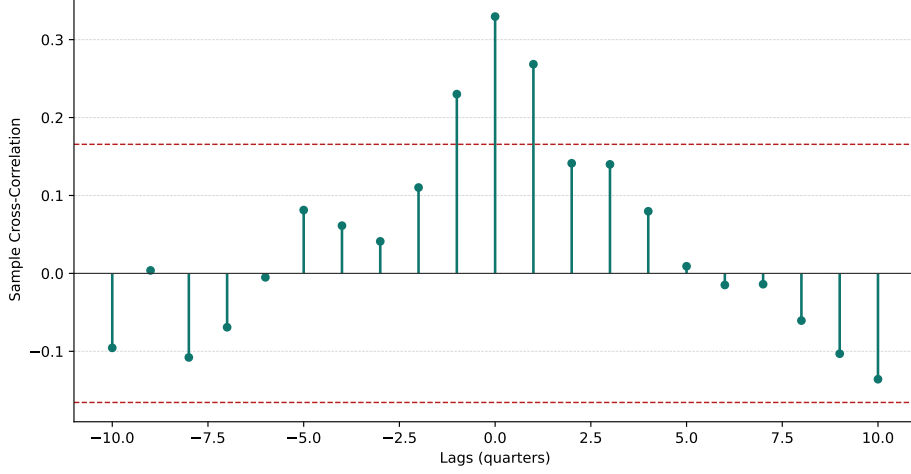


Fig. 4: Sample cross-correlation between quarterly FCIX_t and MFI_t over the period January 1990–December 2024.

contemporaneous, with a peak correlation of approximately 0.33 at lag zero, indicating a moderate near-term alignment between price-based volatility and variation in accounting fundamentals.

While correlation provides a preliminary indication of synchrony, it is limited to capturing linear dependence and cannot rule out the possibility of spurious alignment. To test for general dependence, including nonlinear and higher-order structures, we employ a nonparametric framework based on empirical distribution comparisons. Specifically, we adopt the consistent testing methodology proposed in [3, 4], which compares the joint empirical distribution with the product of its marginals using two divergence-based test statistics.

Let $\{(\text{FCIX}_t, \text{MFI}_t)\}_{t=1}^n$ be a sequence of i.i.d. observations drawn from an unknown joint distribution $\mathbb{G}_{\text{FCIX}, \text{MFI}}$. Denote the corresponding marginal distributions by $\mathbb{G}_{\text{FCIX}}$ and $\mathbb{G}_{\text{MFI}}$. The null hypothesis of independence is expressed as:

$$H_0^{\text{INDP}} : \mathbb{G}_{\text{FCIX}, \text{MFI}} = \mathbb{G}_{\text{FCIX}} \times \mathbb{G}_{\text{MFI}}.$$

Let $\mathcal{P}_n = \{A_{n,1}, \dots, A_{n,m_n}\}$ and $\mathcal{Q}_n = \{B_{n,1}, \dots, B_{n,m'_n}\}$ denote measurable partitions of $\mathbb{R}_{\geq 0}$, constructed to discretize the support of the empirical distributions. The first test statistic is based on the L_1 distance between the joint and product measures:

$$L_n := \sum_{A \in \mathcal{P}_n} \sum_{B \in \mathcal{Q}_n} |\mathbb{G}_{n, \text{FCIX}, \text{MFI}}(A \times B) - \mathbb{G}_{n, \text{FCIX}}(A) \mathbb{G}_{n, \text{MFI}}(B)|. \quad (17)$$

The empirical probabilities are defined by:

$$\mathbb{G}_{n, \text{FCIX}, \text{MFI}}(A \times B) := \frac{1}{n} \sum_{t=1}^n \mathbb{I}[(\text{FCIX}_t, \text{MFI}_t) \in A \times B], \quad (18)$$

$$\mathbb{G}_{n,\text{FCIX}}(A) := \frac{1}{n} \sum_{t=1}^n \mathbb{I}[\text{FCIX}_t \in A], \quad (19)$$

$$\mathbb{G}_{n,\text{MFI}}(B) := \frac{1}{n} \sum_{t=1}^n \mathbb{I}[\text{MFI}_t \in B]. \quad (20)$$

As an alternative, we compute a log-likelihood divergence statistic, which measures the Kullback–Leibler distance between the empirical joint and product-marginal distributions:

$$I_n := \sum_{A \in \mathcal{P}_n} \sum_{B \in \mathcal{Q}_n} \mathbb{G}_{n,\text{FCIX},\text{MFI}}(A \times B) \log \left(\frac{\mathbb{G}_{n,\text{FCIX},\text{MFI}}(A \times B)}{\mathbb{G}_{n,\text{FCIX}}(A) \mathbb{G}_{n,\text{MFI}}(B)} \right). \quad (21)$$

Under the null hypothesis of independence, both L_n and I_n converge in distribution, and critical values can be approximated either via permutation or using asymptotic theory.

Table 2 reports the results of both tests applied to $\{\text{FCIX}_t\}_{t=1}^{140}$ and $\{\text{MFI}_t\}_{t=1}^{140}$, using $m_n = m'_n = 6$ equal-frequency discretization bins per dimension. Critical thresholds are obtained from 10,000 random permutations of the paired quarterly series.

Table 2: Results of independence tests between quarterly FCIX_t and MFI_t .

	L_1 -based	Log-likelihood
Test statistic	0.544	0.242
Critical thresholds		
1% level	0.459	0.175
5% level	0.420	0.147
10% level	0.403	0.134

Both test statistics exceed their respective critical values at the 1% significance level, leading to the rejection of the null hypothesis H_0^{INDP} . This provides strong statistical evidence that FCIX_t and MFI_t are dependent, indicating that price-based market volatility and structural shifts in accounting fundamentals are jointly distributed and possibly mutually informative. These findings serve as a critical empirical foundation for evaluating market efficiency in the subsequent section.

4 Predictive Modeling for Fundamentals

Having established the MFI as a summary of cross-sectional dispersion in fundamentals and demonstrated its statistical dependence with FCIX, we now assess whether the underlying tensor structure contains predictive information for next-quarter accounting values. The task is to forecast a firm-by-feature target matrix using

the preceding L quarters of fundamentals, with $L \in \{2, 4\}$ corresponding to 6-month and 1-year lookbacks.

4.1 Tensor construction and imputation

From the 40 fundamental features identified in Section 2.1, we construct a quarterly predictive panel using the same cleaned Compustat extract as the MFI analysis. Year-to-date cash-flow fields are converted to quarterly flows by within-fiscal-year differencing, while genuinely quarterly and trailing-twelve-month fields are retained as reported. The prediction universe consists of the top $N = 50$ firms by Compustat market value ('mkvalqt') at 2024Q4, which provides a reproducible large-cap universe while keeping the tensor-regression problem computationally tractable. We observe these firms over 2005Q1–2024Q4, snapping report dates to a strict quarterly grid and keeping the last report per firm-quarter. This produces a raw tensor with shape $50 \times 40 \times 80$ and observed-entry density approximately 90.3%. Each feature is transformed by the log-modulus map

$$\phi(x) = \text{sgn}(x) \log(1 + |x|), \quad (22)$$

which stabilizes scale while preserving sign information. The resulting raw tensor is $\mathcal{X} \in \mathbb{R}^{N \times F \times T}$.

For each time index t , we form an input window and target:

$$\mathcal{X}_t = \mathcal{X}_{:, :, t:t+L-1} \in \mathbb{R}^{N \times F \times L}, \quad \mathbf{Y}_t = \mathcal{X}_{:, :, t+L} \in \mathbb{R}^{N \times F}, \quad (23)$$

with observation masks $\mathcal{M}_t$ and $\mathbf{M}_t$ for the input and target, respectively. Stacking across all valid time indices yields a four-dimensional input tensor $\mathcal{X}_{\text{all}} \in \mathbb{R}^{T' \times N \times F \times L}$ and a three-dimensional target $\mathbf{Y}_{\text{all}} \in \mathbb{R}^{T' \times N \times F}$.

Missing entries inside each input window are imputed with a mask-aware Tucker decomposition,

$$\min_{\mathcal{H}_t, \mathbf{A}_t, \mathbf{B}_t, \mathbf{C}_t} \|\mathcal{M}_t \odot (\mathcal{X}_t - \mathcal{H}_t \times_1 \mathbf{A}_t \times_2 \mathbf{B}_t \times_3 \mathbf{C}_t)\|_F^2, \quad (24)$$

where $\mathcal{M}_t$ restricts the objective to observed entries. The fitted Tucker tensor is used only to fill missing entries; observed accounting values are preserved exactly in the covariate tensor passed to the regression step. This distinction is important because the imputer is intended to complete sparse fundamentals, not to smooth or denoise reported values.

The Tucker ranks are selected by a held-out-entry validation procedure. For each rolling window, we hide 10% of observed entries, stratified by feature, fit the mask-aware Tucker model on the remaining observed entries, and score relative Frobenius error only on the hidden entries. The time-mode rank is fixed at L , and the firm- and feature-mode ranks are swept over a small grid. This procedure selects ranks $(2, 2, 2)$ for $L = 2$ and $(4, 4, 4)$ for $L = 4$; the $L = 2$ choice is the parsimonious one-standard-error selection, while $(4, 4, 4)$ minimizes the validation error for the one-year lookback. The final imputation step uses SVD initialization, a maximum of 100

iterations, and tolerance 10^{-5} . Figure 5 illustrates the decomposition structure. We compute an observed-only RMS scale for each window,

$$r_t = \sqrt{\frac{1}{|\Omega_t|} \sum_{(i,j,k) \in \Omega_t} x_{ijk}^2}, \quad (25)$$

where Ω_t is the set of observed entries in $\mathcal{X}_t$. We evaluate two input-scaling variants: an unscaled covariate tensor in which missing entries are imputed in original units and observed entries remain unchanged (denoted Unscaled), and an RMS-normalized covariate tensor in which the completed input window is divided by r_t (denoted Normalized).

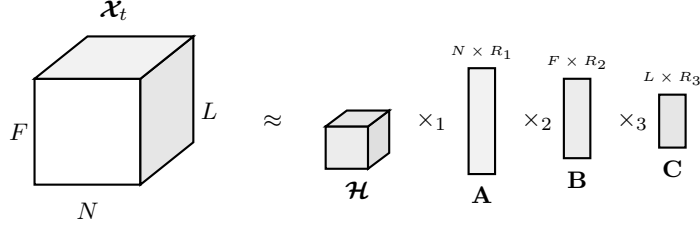


Fig. 5: Mask-aware Tucker imputation for sparse fundamentals. To resolve missing entries without discarding sparse features, the incomplete input tensor $\mathcal{X}_t \in \mathbb{R}^{N \times F \times L}$ is approximated by a dense core tensor $\mathcal{H} \in \mathbb{R}^{R_1 \times R_2 \times L}$ contracted with orthogonal factor matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ along each respective mode. The validated ranks are $(R_1, R_2, L) = (2, 2, 2)$ for the two-quarter lookback and $(4, 4, 4)$ for the four-quarter lookback.

4.2 Low-rank CP regression

To stabilize the target tensor prior to estimation, we apply a within-firm centering analogous to fixed effects in panel econometrics. For each firm–feature pair, we compute a mask-aware temporal mean:

$$\mu_{ij} = \frac{\sum_t m_{tij} y_{tij}}{\sum_t m_{tij} + \varepsilon}, \quad (26)$$

where m_{tij} is the observation indicator and $\varepsilon = 10^{-8}$ prevents division by zero. The centered residuals are further normalized by their observed root-mean-square,

$$s = \sqrt{\frac{1}{|\Omega|} \sum_{(t,i,j) \in \Omega} (y_{tij} - \mu_{ij})^2}, \quad (27)$$

where $\Omega = \{(t, i, j) : m_{tij} = 1\}$, yielding the scaled target $\tilde{y}_{tij} = m_{tij}(y_{tij} - \mu_{ij})/s$. Missing entries are set to zero in the scaled target.

We model the scaled, de-meaned target with a low-rank CANDECOMP/PARAFAC (CP) tensor regression. The coefficient tensor $\mathcal{W} \in \mathbb{R}^{N \times F \times L \times N \times F}$ admits the CP form

$$\mathcal{W} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r \circ \mathbf{d}_r \circ \mathbf{e}_r, \quad (28)$$

where $\mathbf{a}_r \in \mathbb{R}^N$, $\mathbf{b}_r \in \mathbb{R}^F$, and $\mathbf{c}_r \in \mathbb{R}^L$ correspond to the input modes, while $\mathbf{d}_r \in \mathbb{R}^N$ and $\mathbf{e}_r \in \mathbb{R}^F$ correspond to the output modes of the r -th rank-one component. The prediction at time t is obtained by inverting the centering and scaling:

$$\hat{\mathbf{Y}}_t = \boldsymbol{\mu} + s \cdot \langle \mathcal{X}_t, \mathcal{W} \rangle \in \mathbb{R}^{N \times F}, \quad (29)$$

where $\boldsymbol{\mu} \in \mathbb{R}^{N \times F}$ is the firm-feature mean matrix and s is the RMS scale defined above. Figure 6 illustrates the structure of the CP regression model.

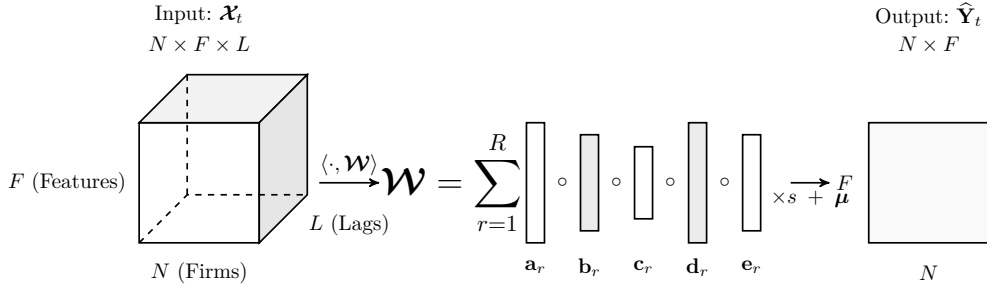


Fig. 6: Low-rank CP tensor regression model. The 5D weight tensor $\mathcal{W}$ is parameterized as a sum of R rank-one components. The multilinear contraction $\langle \mathcal{X}_t, \mathcal{W} \rangle$ maps the historical 3D input ($N \times F \times L$) to a 2D next-quarter prediction ($N \times F$), which is then rescaled by s and shifted by the firm-feature mean matrix $\boldsymbol{\mu}$.

We estimate $\mathcal{W}$ via TensorLy [5] by regularized least squares,

$$\min_{\mathcal{W}} \sum_t \left\| \tilde{\mathbf{Y}}_t - \langle \mathcal{X}_t, \mathcal{W} \rangle \right\|_F^2 + \lambda \|\mathcal{W}\|_F^2, \quad (30)$$

where $\tilde{\mathbf{Y}}_t$ is the zero-filled, centered, RMS-scaled target defined above. Because TensorLy's CP solver does not accept observation masks, missing entries are set to zero prior to fitting; all evaluation at test time remains fully mask-aware. Hyperparameters—the rank $R \in \{5, \dots, 50\}$ and regularization weight $\lambda \in [10^{-4}, 100]$ —are tuned via Optuna on a rolling three-fold time-series split of the development data (80% of windows).

4.3 Ridge baseline and evaluation

As a competitive baseline, we fit per-feature ridge regressions on vectorized lag windows. Both the CP and ridge models share the same within-firm centering described in Section 4.2; the key distinction is that each ridge model is fitted independently for a single feature with its own regularization penalty, whereas the CP tensor shares one low-rank parameterization across all features simultaneously. For each feature $j \in \{1, \dots, F\}$, let $\mathbf{y}^{(j)} \in \mathbb{R}^{T'N}$ denote the stacked target vector and $\mathbf{X} \in \mathbb{R}^{T'N \times FL}$ the design matrix formed by flattening the lag dimension. The per-feature model is

$$\mathbf{y}^{(j)} = \mathbf{X}\boldsymbol{\beta}^{(j)} + \boldsymbol{\mu}^{(j)} + \boldsymbol{\epsilon}^{(j)}, \quad (31)$$

where $\boldsymbol{\mu}^{(j)}$ contains the firm fixed effects for feature j .

Our pipeline computes mask-aware firm-feature means $\mu_i^{(j)}$ and models the de-meaned target directly—the classical within-transformation in panel econometrics—removing the fixed effects before estimation and restoring them at prediction time.

To make the panel structure explicit, let $\mathbf{y}_i^{(j)} \in \mathbb{R}^{T'}$ denote the time-series target for firm i and feature j , and let $\mathbf{X}_i \in \mathbb{R}^{T' \times FL}$ denote the firm’s design matrix. The stacked panel regression groups observations by firm:

$$\begin{bmatrix} \mathbf{y}_1^{(j)} \\ \mathbf{y}_2^{(j)} \\ \vdots \\ \mathbf{y}_N^{(j)} \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_N \end{bmatrix} \boldsymbol{\beta}^{(j)} + \begin{bmatrix} \boldsymbol{\mu}_1^{(j)} \\ \boldsymbol{\mu}_2^{(j)} \\ \vdots \\ \boldsymbol{\mu}_N^{(j)} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\epsilon}_1^{(j)} \\ \boldsymbol{\epsilon}_2^{(j)} \\ \vdots \\ \boldsymbol{\epsilon}_N^{(j)} \end{bmatrix}, \quad (32)$$

where $\boldsymbol{\mu}_i^{(j)} \in \mathbb{R}^{T'}$ replicates firm i ’s historical mean across every time step. For a generic firm i , the internal elements of $\mathbf{y}_i^{(j)}$ and $\mathbf{X}_i$ expand across time steps, features, and lags as:

$$\mathbf{y}_i^{(j)} = \begin{bmatrix} y_{i,t+1}^{(j)} \\ \vdots \\ y_{i,T'}^{(j)} \end{bmatrix}, \quad \mathbf{X}_i = \begin{bmatrix} \overbrace{x_{i,t}^{(1)} \cdots x_{i,t}^{(F)}}^{\text{Lag 1}} & \cdots & \overbrace{x_{i,t-L+1}^{(1)} \cdots x_{i,t-L+1}^{(F)}}^{\text{Lag L}} \\ \vdots & \ddots & \vdots \\ x_{i,T'-1}^{(1)} \cdots x_{i,T'-1}^{(F)} & \cdots & x_{i,T'-L}^{(1)} \cdots x_{i,T'-L}^{(F)} \end{bmatrix}. \quad (33)$$

To ensure a rigorously competitive benchmark, ridge penalties are independently optimized for each feature via an inner three-fold time-series cross-validation from the candidate set $\alpha \in \{10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4\}$. This guarantees that the baseline is optimally tuned for the specific scale and variance of every individual fundamental feature. When a feature has too few observations (below an adaptive threshold), we revert to predicting the firm-mean baseline.

Performance is evaluated using a pooled, mask-aware coefficient of determination,

$$R^2 = 1 - \frac{\sum_{t,i,j} m_{tij} (y_{tij} - \hat{y}_{tij})^2}{\sum_{t,i,j} m_{tij} (y_{tij} - \bar{y})^2}, \quad (34)$$

with $\bar{y}$ the mean of observed entries. We also compute per-feature R^2 values to identify which fundamentals are most predictable. To assess whether CP regression systematically outperforms ridge, we estimate AR(1) persistence per feature and examine the association between persistence and the performance gap ($\Delta = R_{\text{CP}}^2 - R_{\text{Ridge}}^2$) using Spearman correlation and paired Wilcoxon tests with Benjamini–Hochberg adjustment for multiple comparisons.

4.4 Empirical results

We now present the comparative forecasting performance of the CP tensor regression against the highly structured ridge baseline. Table 3 reports the pooled, mask-aware coefficient of determination (R^2) on the held-out test (20%) set for each input scaling mode (Unscaled and Normalized) and lookback horizon ($L \in \{2, 4\}$). Both models share the same within-firm centering (Section 4.2), but the ridge baseline enjoys two structural advantages: (i) a separate, independently fitted model for every feature, whereas the tensor shares a single low-rank parameterization across all features simultaneously; and (ii) per-feature regularization penalties selected by inner cross-validation, whereas the tensor uses one global set of hyperparameters. Despite these advantages, the CP tensor matches or exceeds the ridge baseline across all configurations, demonstrating that the cross-feature multilinear structure it captures is genuinely predictive and cannot be replicated by even a well-tuned collection of independent regressions. To further understand where the tensor formulation provides the most value, Table 4 summarizes the feature-level AR(1) persistence analysis, quantifying the statistical significance of the performance gap (Δ) using Benjamini–Hochberg adjusted p -values.

Table 3 shows that the CP tensor regression outperforms the ridge baseline across all four configurations, with the largest gains emerging under Normalized scaling at $L = 4$ ($\Delta = +0.028$). The consistent positive deltas demonstrate that the multilinear structure captured by the 5D weight tensor provides genuine predictive value beyond what per-feature linear models can achieve. Notably, the Normalized mode yields the strongest CP performance overall (Test $R^2 = 0.784$ at $L = 4$), confirming that RMS normalization removes magnitude-driven scaling artifacts and allows the tensor to focus on the underlying cross-firm, cross-feature dynamics.

Table 4 confirms that the CP tensor provides a statistically significant predictive advantage over the ridge baseline in three of four configurations (Wilcoxon $p_{\text{adj}} < 0.01$). The relationship between AR(1) persistence and the performance gap is mode-dependent: under Normalized scaling, the CP advantage concentrates on more persistent features ($r = 0.468$, $p_{\text{adj}} = 0.042$ at $L = 2$), while under Unscaled scaling at $L = 4$ the pattern reverses sharply ($r = -0.835$, $p_{\text{adj}} < 10^{-5}$), with the tensor excelling on less persistent series. This mode dependence suggests that RMS normalization fundamentally alters which cross-feature dynamics the tensor exploits,

Table 3: CP tensor regression versus ridge baseline: pooled R^2 on held-out test (20%) set.

Mode	L	Ridge Test R^2	CP Test R^2	Δ
Unscaled	2	0.760	0.778	+0.018
Unscaled	4	0.758	0.766	+0.008
Normalized	2	0.750	0.768	+0.018
Normalized	4	0.756	0.784	+0.028

and that the multilinear structure captures predictive information inaccessible to per-feature regressions regardless of the persistence regime.

Table 4: Persistence diagnostics and statistical significance (Benjamini–Hochberg adjusted p -values).

Mode	L	Mean Δ	Spearman r	Spearman p_{adj}	Wilcoxon p_{adj}
Unscaled	2	+0.152	0.395	0.075	0.0002
Unscaled	4	+0.030	−0.835	0.000002	0.197
Normalized	2	+0.168	0.468	0.042	0.002
Normalized	4	+0.124	0.054	0.802	0.0000005

5 Conclusion

In this paper, we introduced a novel, tensor-based framework for modeling and forecasting firm-level fundamentals. By constructing the Market Fundamentals Index (MFI), we provided a structural lens into market-wide volatility, successfully demonstrating that fluctuations in accounting fundamentals are statistically dependent on price-based chaos (FCIX).

Building upon this structural relationship, we developed a rigorous predictive pipeline that resolves real-world data sparsity via mask-aware Tucker imputation. We demonstrated that a low-rank CANDECOMP/PARAFAC (CP) tensor regression—when properly controlled with firm-by-feature fixed effects—can effectively exploit the multilinear dynamics between historical lags, firm heterogeneity, and cross-feature interactions. Ultimately, this framework establishes a robust foundation for systematically evaluating the predictability of quarterly fundamentals, paving the way for future extensions incorporating broader asset universes and macroeconomic covariates.

References

- [1] Frey, B.J., Dueck, D.: Clustering by passing messages between data points. *Science*, 315(5814), 972–976 (2007)
- [2] Dueck, D.: Affinity propagation: Clustering data by passing messages. PhD thesis, University of Toronto (2009)
- [3] Gretton, A., Györfi, L.: Nonparametric independence tests: Space partitioning and kernel approaches. In: *International Conference on Algorithmic Learning Theory*, pp. 183–198 (2008). Springer
- [4] Gretton, A., Györfi, L.: Consistent nonparametric tests of independence. *Journal of Machine Learning Research*, 11(46), 1391–1423 (2010)
- [5] Kossaifi, J., Panagakis, Y., Anandkumar, A., Pantic, M.: Tensorly: Tensor learning in Python. *Journal of Machine Learning Research*, 20(26), 1–6 (2019)